

Department of Applied Mathematics and Statistics
COLORADO SCHOOL OF MINES
MATH 455: Partial Differential Equations

Let

$$\sin(x + y) = \cos(xy)$$

Note that

$$\cos(xy) = \sin\left(\frac{\pi}{2} - xy\right)$$

so our implicit equation becomes

$$\sin(x + y) = \sin\left(\frac{\pi}{2} - xy\right)$$

Also note that the general solutions to

$$\sin(A) = \sin(B)$$

are

$$A = B + 2\pi k$$

and

$$A = \pi - B + 2\pi k$$

for $k \in \mathbb{Z}$. Then, in our case,

$$\begin{aligned} x + y &= \frac{\pi}{2} - xy + 2\pi k \\ \implies x + y + xy &= \frac{\pi}{2} + 2\pi k \end{aligned}$$

and

$$\begin{aligned} x + y &= \pi - \left(\frac{\pi}{2} - xy\right) + 2\pi k \\ \implies x + y - xy &= \frac{\pi}{2} + 2\pi k \end{aligned}$$

.

So our curve is the union of the two families of curves:

$$x + y + xy = R_k$$

$$x + y - xy = R_k$$

for $k \in \mathbb{Z}$ and $R_k = \frac{\pi}{2} + 2\pi k$

For shorthand, let $P = x + y + xy$ and $Q = x + y - xy$. At the origin $P = Q = 0$, so the closed region containing the origin is bounded by the level curves nearest to zero: $R_{-1} = \frac{-3\pi}{2}$ and $R_0 = \frac{\pi}{2}$:

$$\frac{-3\pi}{2} \leq P, Q \leq \frac{\pi}{2}$$

Over both families of curves, we find the following nearest-zero level curves:

$$P = \frac{-3\pi}{2} \quad (1)$$

$$P = \frac{\pi}{2} \quad (2)$$

$$Q = \frac{-3\pi}{2} \quad (3)$$

$$Q = \frac{\pi}{2} \quad (4)$$

We find pairwise intersections and integrate accordingly. (1) intersects (2) nowhere (for any values of x and y , the expression $P(x, y)$ can't be equal to both $-3\pi/2$ and $\pi/2$). Similarly, (3) intersects (4) nowhere.

Intersection of (1) and (3):

$$x + y + xy = \frac{-3\pi}{2}$$

$$x + y - xy = \frac{-3\pi}{2}$$

Add and subtract the equations to obtain a simpler system:

$$x + y = \frac{-3\pi}{2} \quad xy = 0$$

Which has solutions

$$\left(0, \frac{-3\pi}{2}\right) \quad \left(\frac{-3\pi}{2}, 0\right)$$

We apply an identical strategy, and find the intersection points of (2) and (4) to be

$$\left(0, \frac{\pi}{2}\right) \quad \left(\frac{\pi}{2}, 0\right)$$

Proceeding identically, we find the intersection points of (1) and (4) to satisfy the quadratic

$$x^2 + \frac{\pi}{2}x - \pi = 0$$

Let a be the positive solution

$$a = \frac{-\pi + \sqrt{\pi^2 + 16\pi}}{4}$$

and similarly let b be the negative solution

$$b = \frac{-\pi - \sqrt{\pi^2 + 16\pi}}{4}$$

Then our intersection points are

$$\left(a, -\frac{\pi}{2} - a\right) \quad \left(b, -\frac{\pi}{2} - b\right)$$

For curves (2) and (3) we obtain a very similar quadratic equation that has no solution due to sign differences. There are no more pairwise intersections to check; hence we've found all of them. Ordered by x -coordinate, we have intersections at

$$x_1 = L = \frac{-3\pi}{2} \approx -4.71238898$$

$$x_2 = b \approx -2.72406857$$

$$x_3 = 0 \quad (\text{repeated})$$

$$x_4 = a \approx 1.15327223888$$

$$x_5 = U = \frac{\pi}{2} \approx 1.57079633$$

(Let L and U be the lower and upper bounds, respectively). We find pairs of vertically-bounding curves: curves (1) and (3) bound between $x = L$ and $x = b$. Curves (3) and (4) bound between $x = b$ and $x = 0$. Curves (1) and (2) bound between $x = 0$ and $x = a$. Curves (2) and (4) bound between $x = a$ and $x = U$.

Because curves (1), (2), (3), (4) are hyperbolas, each can be written as a function for the sake of the integration. For each of our previous 4 curves, we solve for y , yielding:

$$y = \frac{L - x}{1 + x} \quad (1)$$

$$y = \frac{U - x}{1 + x} \quad (2)$$

$$y = \frac{L - x}{1 - x} \quad (3)$$

$$y = \frac{U - x}{1 - x} \quad (4)$$

Hence, our bounded area is

$$\begin{aligned} & \int_L^b \left(\frac{L - x}{1 + x} - \frac{L - x}{1 - x} \right) dx \\ & + \int_b^0 \left(\frac{U - x}{1 - x} - \frac{L - x}{1 - x} \right) dx \\ & + \int_0^a \left(\frac{U - x}{1 + x} - \frac{L - x}{1 + x} \right) dx \\ & + \int_a^U \left(\frac{U - x}{1 + x} - \frac{U - x}{1 - x} \right) dx \\ & = 14.7660251359 \\ & = 14.76603 \quad (\text{rounded to 5 decimal places}) \end{aligned}$$